

HARKIRAT SINGH

Generative AI Engineer

Pune, India · harkirat.email@gmail.com · +91 9453675860 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Generative AI Engineer with hands-on experience building LLM-powered applications using LangChain Chains & Runnables, RAG pipelines, and vector databases. Skilled in OpenAI and Anthropic APIs with a strong Python backend foundation in FastAPI and SQLAlchemy. Passionate about designing scalable, production-ready AI solutions with clean, modular, and well-documented code.

TECHNICAL SKILLS

Gen AI: LangChain, LlamaIndex, LCEL Runnables & Chains, RAG Pipelines, Prompt Engineering

LLM APIs: OpenAI API (GPT-4o, Embeddings), Anthropic API (Claude), Hugging Face

Vector DBs: Pinecone, ChromaDB, FAISS

Backend: FastAPI, SQLAlchemy, REST APIs, JWT Authentication, Pydantic, Python

Data & Tools: Pandas, MySQL, SQLite, Scrapy, BeautifulSoup, APScheduler, Git, Logging

PROJECTS

RAG-Powered Document Q&A System *Personal Project · 2026*

- Designed an end-to-end RAG pipeline using LangChain & ChromaDB enabling semantic search across 50+ custom documents, achieving sub-2s average query response time and ~90% answer relevance on internal test sets.
- Integrated OpenAI text-embedding-ada-002 for chunking and retrieval; used GPT-4o via LCEL Chains & Runnables for answer synthesis with context-aware prompting.
- Deployed pipeline behind a FastAPI REST endpoint with JWT-secured access, Pydantic request validation, structured logging, and Docker-ready containerization.

LangChain Agent with Tool Use *Personal Project · 2025*

- Built a multi-turn LCEL agent dynamically routing queries to 3 custom tools (web search, calculator, DB lookup) with persistent conversation memory across sessions.
- Integrated Anthropic Claude API as the primary LLM backbone with automatic fallback to OpenAI GPT-4o for cost optimization — reducing per-query cost by ~40% without degrading quality.
- Designed modular tool-calling architecture with error handling and retry logic, validated across 200+ test queries with reliable multi-step reasoning.

Web Scraping & Data Pipeline Automation *Personal Project · 2025*

- Built a production-grade scraper using Scrapy and BeautifulSoup to extract structured product data from 5+ e-commerce sites, processing 10,000+ records per scheduled run.
- Automated data cleaning, transformation, and storage with Pandas and MySQL; implemented APScheduler-based cron jobs for fully hands-off daily execution with CSV/JSON export support.
- Achieved 98%+ pipeline success rate across 30 days of continuous operation via structured logging, error alerting, and intelligent retry mechanisms.

REST API — Book Inventory System *Freelance · 2024*

- Designed and delivered a FastAPI backend with SQLAlchemy ORM (SQLite/MySQL), JWT authentication, Pydantic validation, auto-generated Swagger UI, and structured logging.
- Reduced client's manual inventory management effort by an estimated 60% by automating CRUD operations and exposing a clean REST-accessible data layer.

EDUCATION

Bachelor of Engineering — Computer Engineering 2023 – 2027

Sinhgad Institute of Technology, Pune